Stephen A. Rice

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SUMMARY

Software Engineer with experience in systems software, distributed backend services, and hardware integration. Built production systems in Rust, Go, and C# for an industrial controls product, owning delivery from design through deployment, and bring CS plus nuclear engineering training to safety-critical, high-reliability work.

EXPERIENCE

Software Engineer · Promess Incorporated *Feb 2021 - May 2025*

- Built Rust and C# device gateway services that handled LAN-based device discovery (UDP broadcast, mDNS/Bonjour), persistent socket connections to industrial press controllers, and bidirectional real-time telemetry streaming to web frontends via gRPC — owning each service from design through release.
- Designed and implemented a runtime behavioral monitoring engine in Rust: ingested expected system behaviors expressed in natural language, derived state subscriptions automatically, and evaluated live event streams against those specifications using a debounced rolling-timeout model to handle out-of-order asynchronous transitions.
- Built an integration testing framework in Rust that exercised the full API surface of a production web application, covering functional correctness and authentication/authorization behavior, with support for serial and parallel tests.
- Developed a machine authorization and traceability backend service in Go: validated barcode-scanned tooling, part, and workstation combinations against a production database, automatically loaded correct part programs on match, and relayed real-time cycle data and events via MQTT.
- Redesigned telemetry pagination for a complex industrial dataset, replacing a full-scan approach with a keyset boundary query that reduced operator-facing load times from several minutes to under one second.
- Contributed backend components to a platform redesign (UltraPro2) including user management, data monitoring, and fieldbus integration, built on a Rust/GraphQL service architecture.
- Built event-driven data pipelines using pub/sub messaging (MQTT, gRPC) supporting ~1k updates/sec, with attention to reliability and predictable behavior under failure conditions.
- Maintained and extended production embedded and desktop software in C#/.NET and C++ for the company's flagship industrial control product (UltraPro).
- Deployed and maintained backend services in containerized Linux environments using Docker; worked within CI/CD pipelines on Azure DevOps.

Graduate Research & Teaching Assistant · Eastern Michigan University *Jan 2019 - Dec 2020*

- Built multi-threaded Python tools for web scraping and static analysis to mine large GitHub corpora for NLP and ML research.
- Tutored students in Java and Python; served as teaching assistant for applied cryptography.

PROJECTS

Awaitguard · GitHub App for Async Rust Code Review *Rust, Tokio, Axum, PostgreSQL*

github.com/grandfoosier/awaitguard

- Built a GitHub App that analyzes PR diffs for async concurrency bugs using scope-aware detectors — tracking brace depth and guard lifetimes to flag lock-across-await, blocking calls, and unbounded spawns only on newly added lines, with source line numbers resolved from unified diff hunk headers.
- Designed a Postgres job queue with SKIP LOCKED dequeuing, a partial unique index to idempotently handle concurrent webhook delivery, exponential backoff retries, and a stuck-job sweeper; optional LLM-enriched findings are cached in Postgres by snippet hash to avoid redundant API calls.

Eventful · Async Event Processing Service *Rust, Tokio, Axum*

github.com/grandfoosier/eventful

- Built an event processing service with explicit state machine transitions, idempotent ingestion, bounded concurrency via Tokio semaphores and worker pools, backpressure handling, retry scheduling, and graceful shutdown.
- Instrumented with Prometheus metrics and structured tracing for production-style observability.

EDUCATION

Masters in Computer Science · Eastern Michigan University

Focus: systems, machine learning, software engineering. Thesis: LLM for punctuation prediction using PyTorch on GCP.

Bachelors in Nuclear Engineering & Radiological Sciences · University of Michigan

Focus: energy systems, risk analysis, computational modeling.

SKILLS

Languages

Rust, Go, C#, Python, C++, SQL

Systems

Event-driven architecture, pub/sub, device integration, real-time pipelines

Protocols & APIs

gRPC, MQTT, GraphQL, REST, protobuf, mDNS

Infrastructure

Linux, Docker, CI/CD, Azure DevOps

Practices

Observability, async systems, distributed messaging, cross-compilation